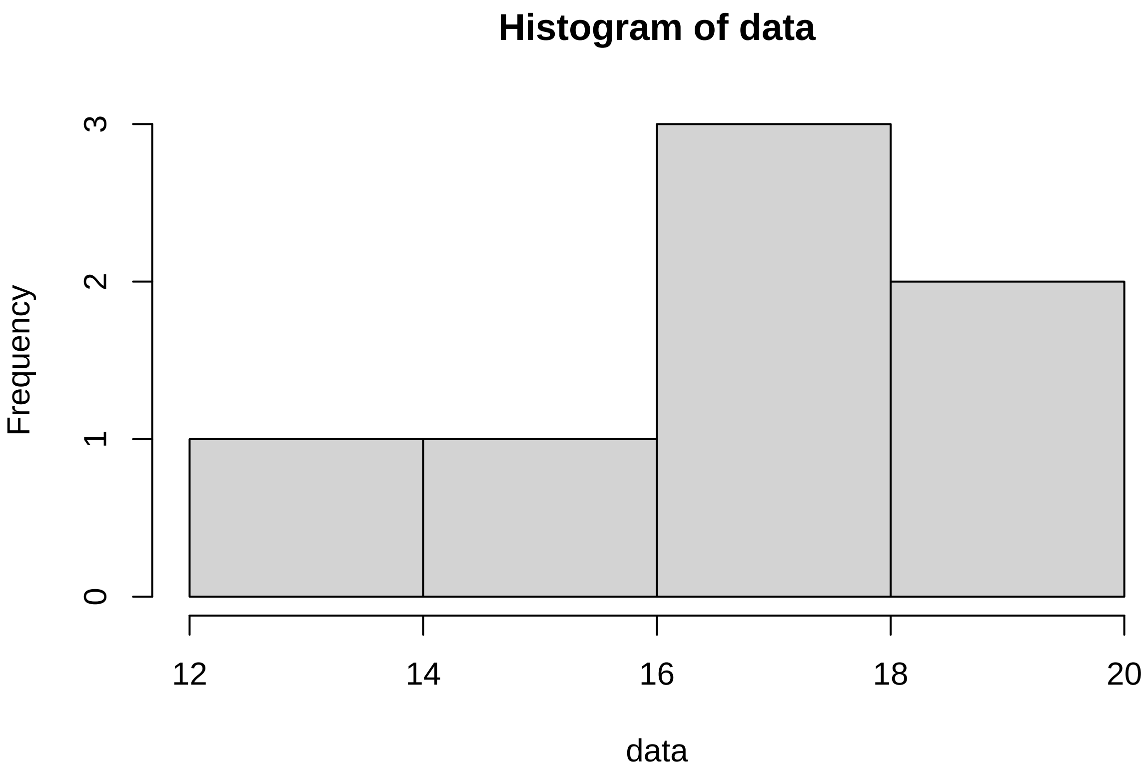
hist.R

macbook

2025-04-12

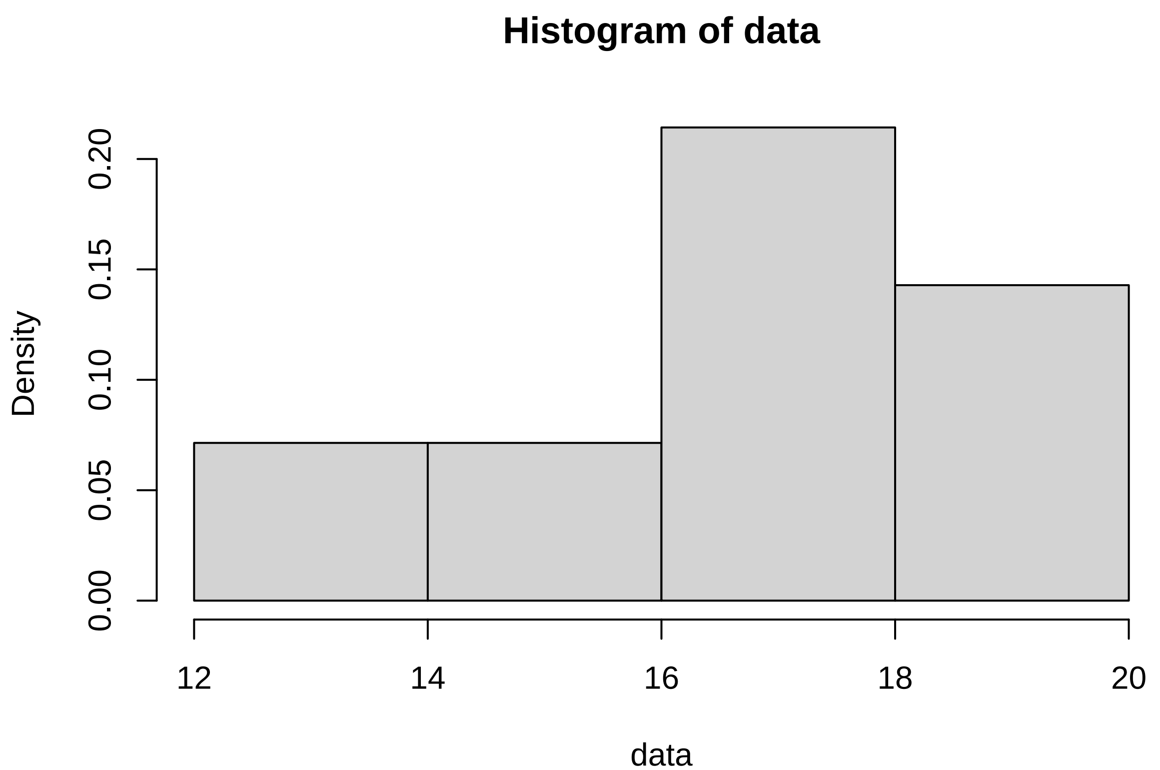
```r
data=c(17,15,12,19,18,20, 17.5)
hist(data)
```

```r
hist(data)
```

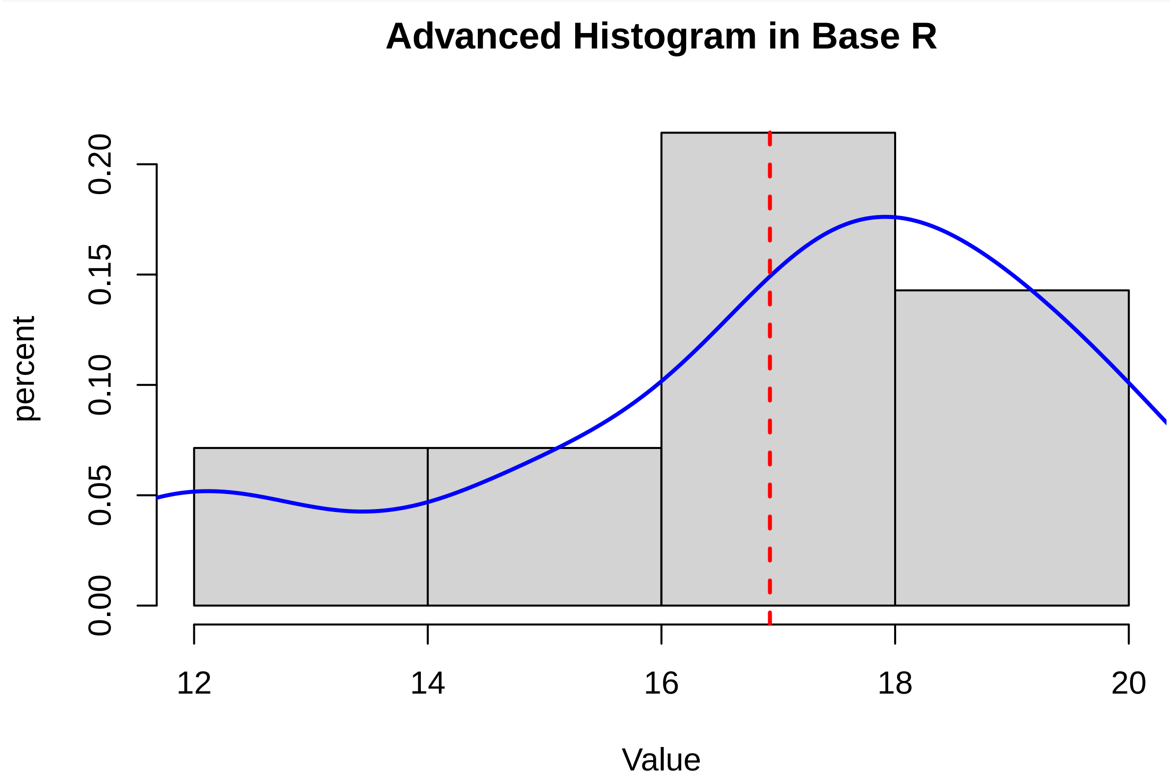



```r
hist(data, breaks = 5)
```

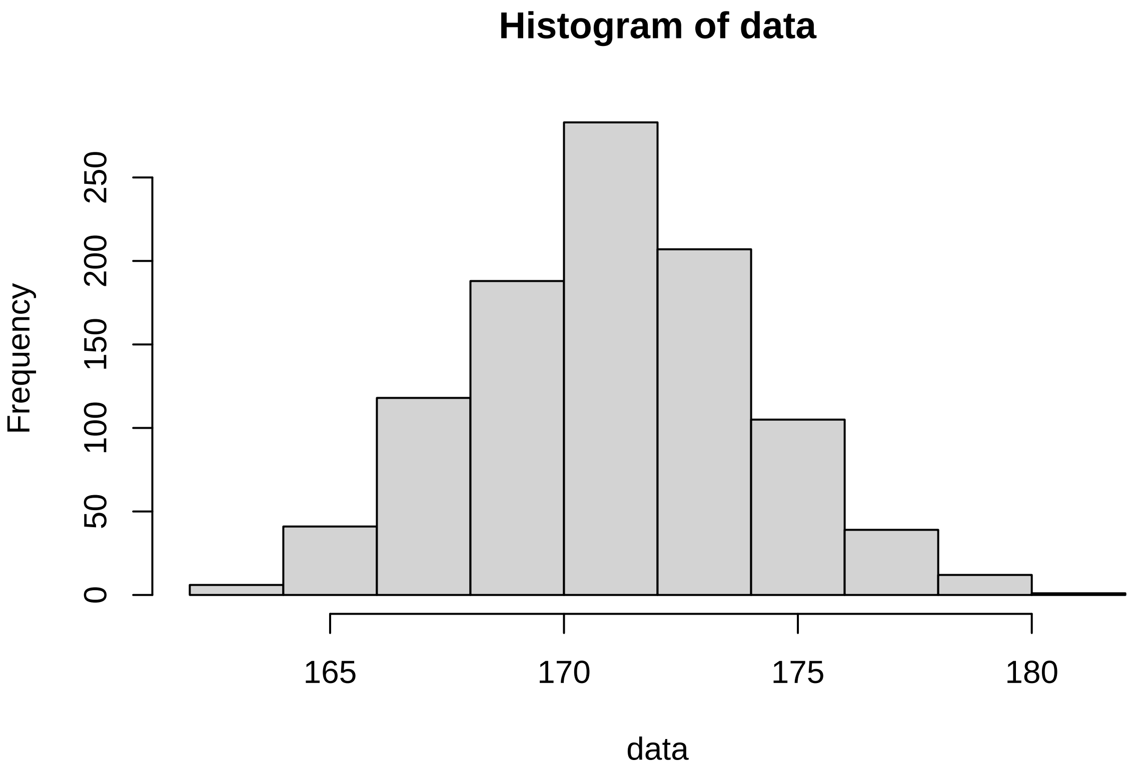
```r
hist(data, probability = T)
```



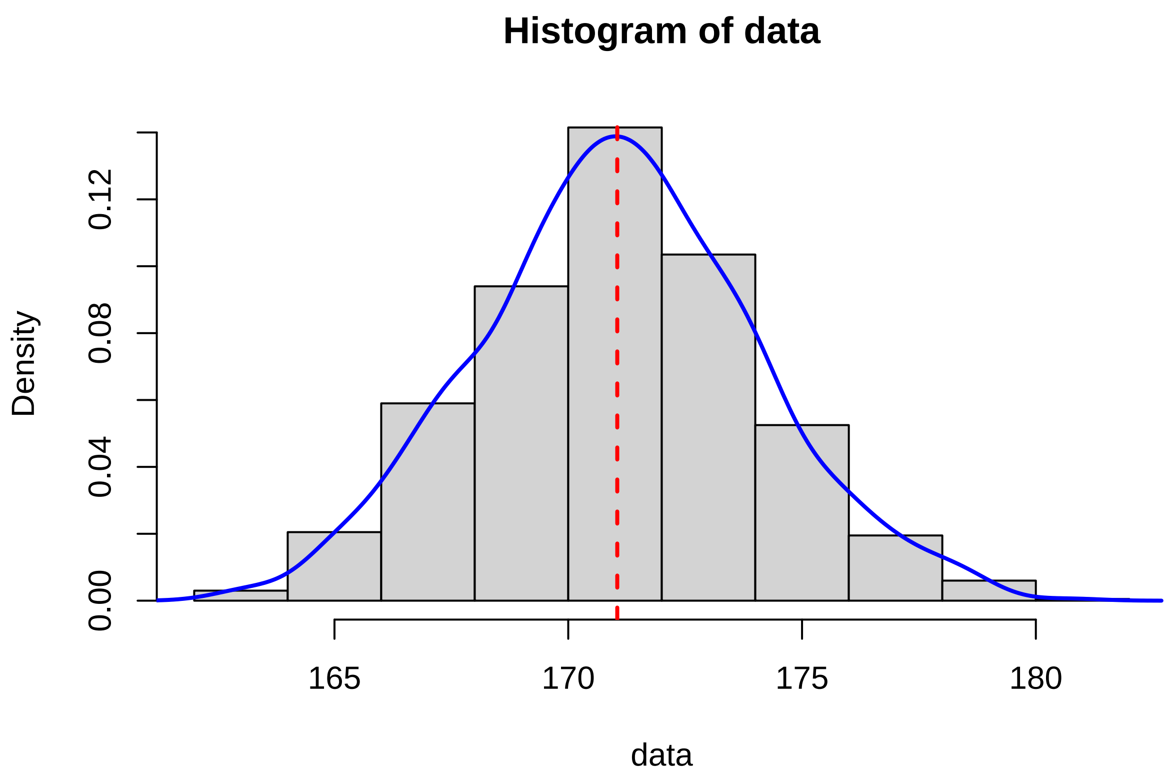

```r
hist(data, probability = T, main = "Advanced Histogram in Base R",
     xlab = "Value",
     ylab="percent")
lines(density(data), col = "blue",lwd=2)
abline(v = mean(data), col = "red", lwd = 2, lty = 2)
```



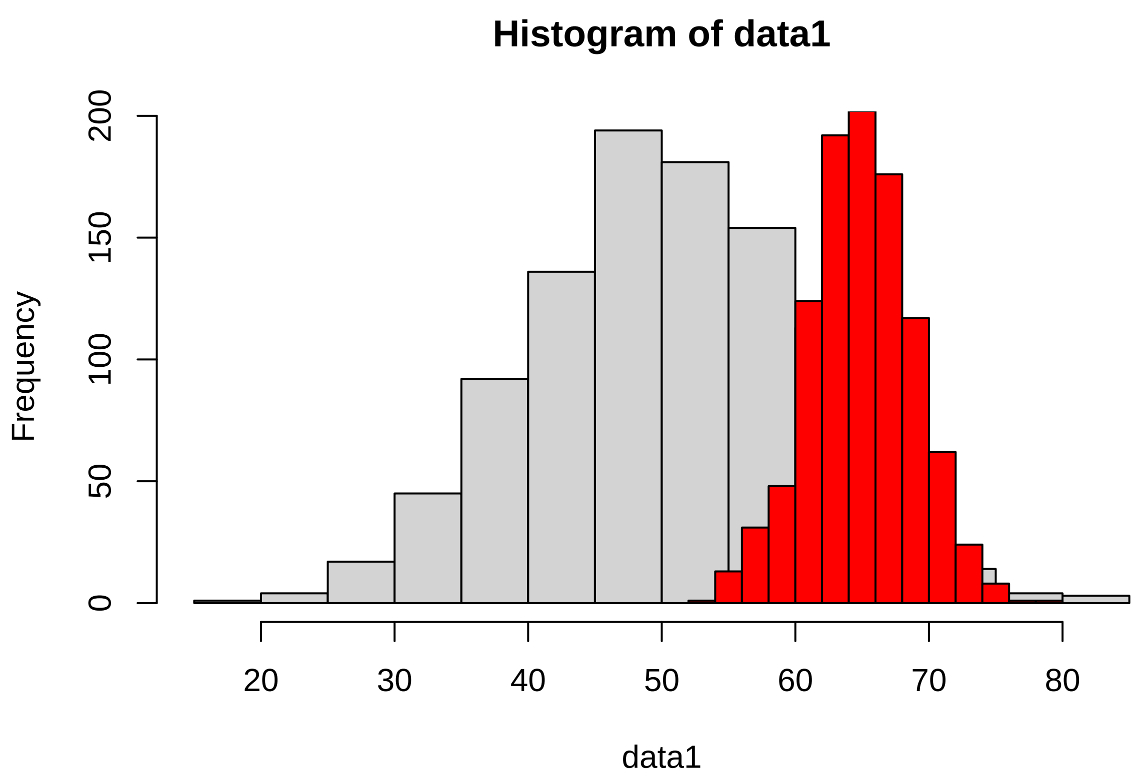
```r
set.seed(123)
data <- rnorm(1000, mean = 171, sd = 3)
hist(data)
```



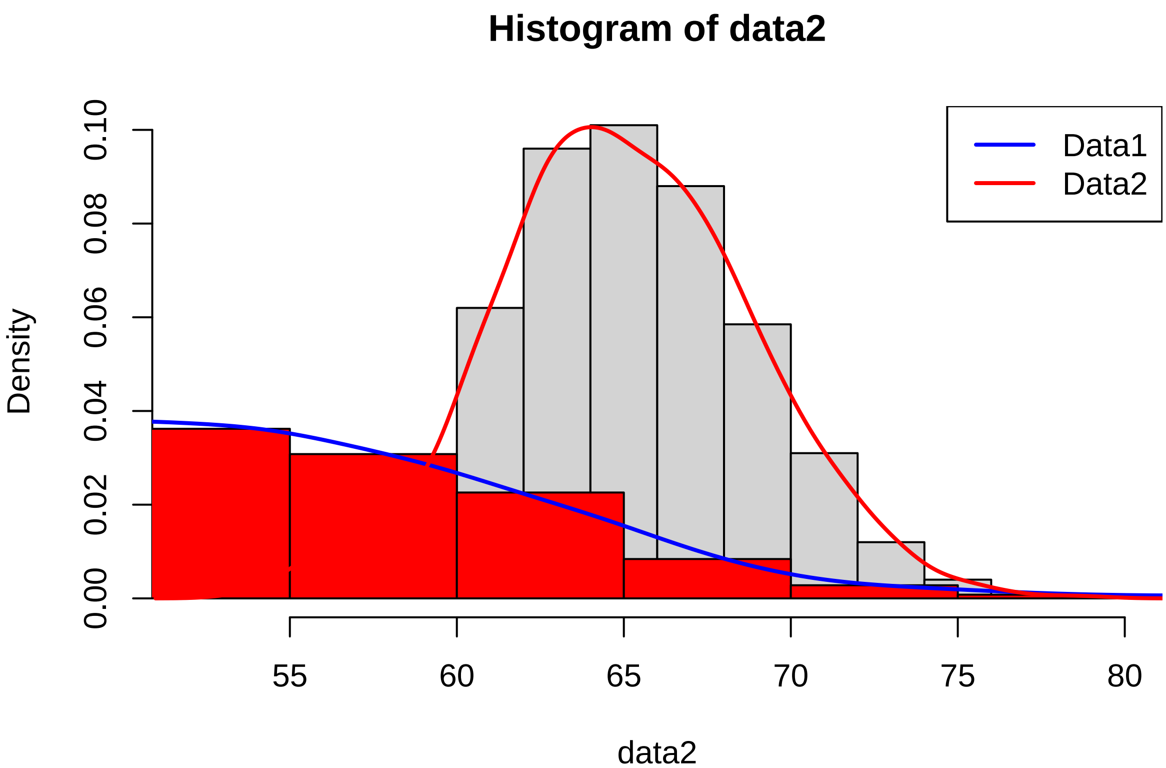

```r
hist(data, probability = T)
lines(density(data), col = "blue",lwd=2)
abline(v = mean(data), col = "red", lwd = 2, lty = 2)
```



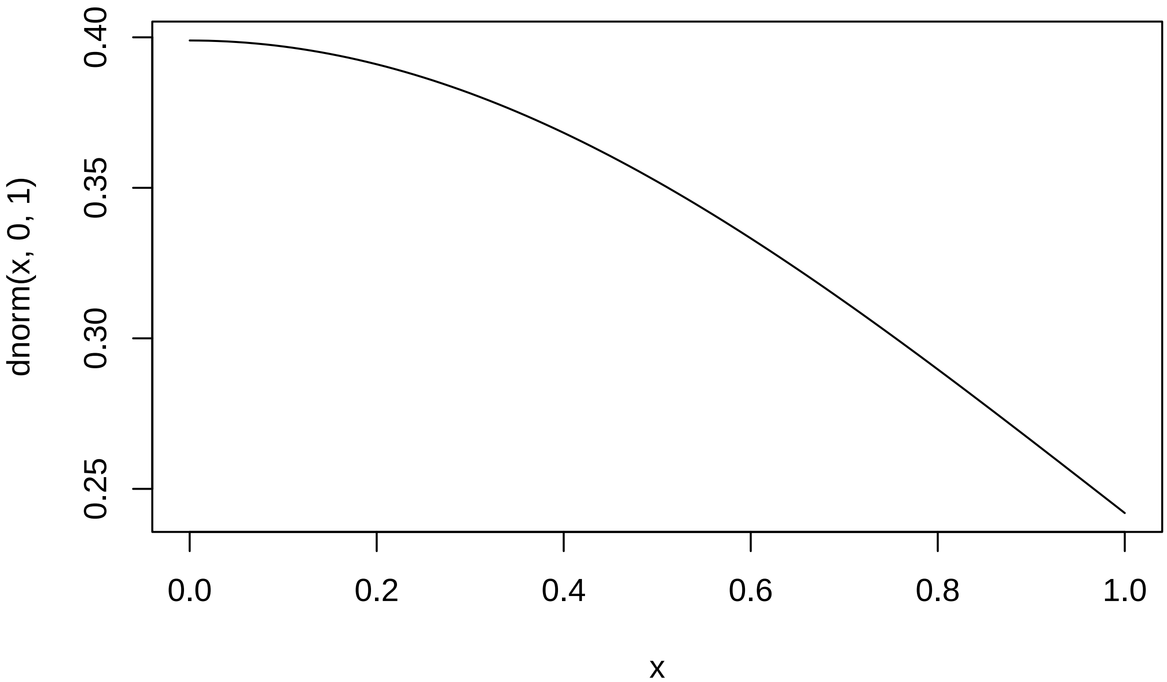

```r
data1 <- rnorm(1000, mean = 50, sd = 10)
data2 <- rnorm(1000, mean = 65, sd = 4)

hist(data1)
hist(data2,add = TRUE, col = 'red')
```



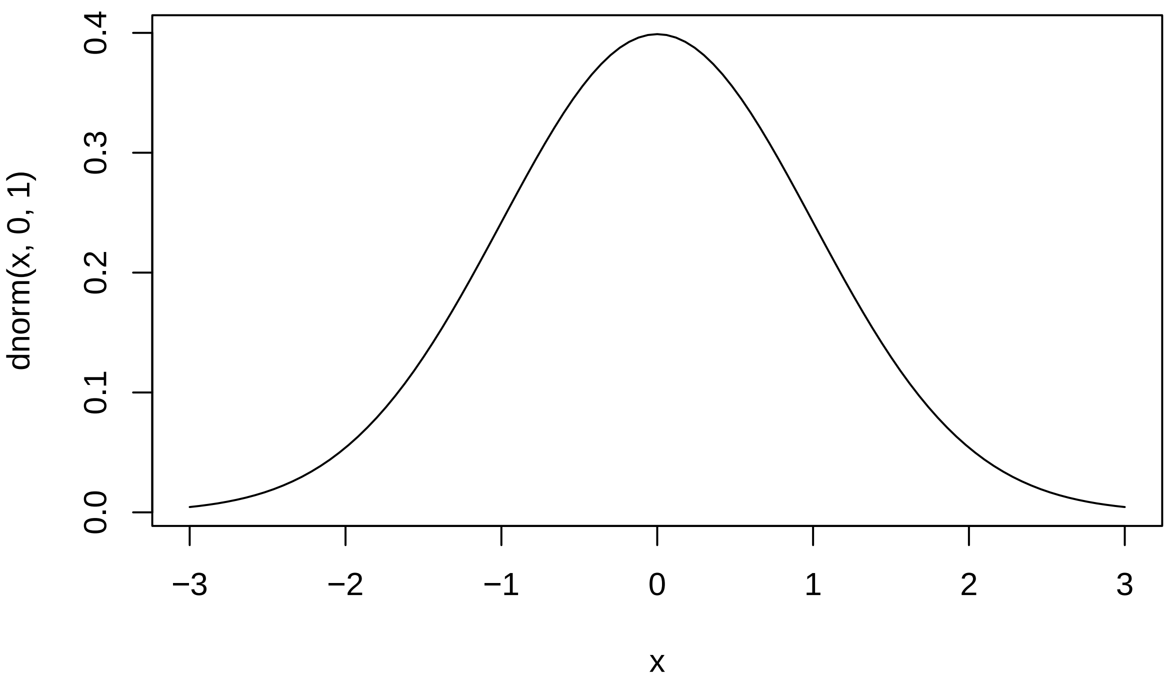
```r
hist(data2, probability = T)
hist(data1,add = TRUE, col = 'red', probability = T)

lines(density(data1), col = "blue", lwd = 2)
lines(density(data2), col = "red", lwd = 2)

legend("topright",
       legend = c("Data1", "Data2"),
       col = c("blue", "red"),
       lwd = 2)
```



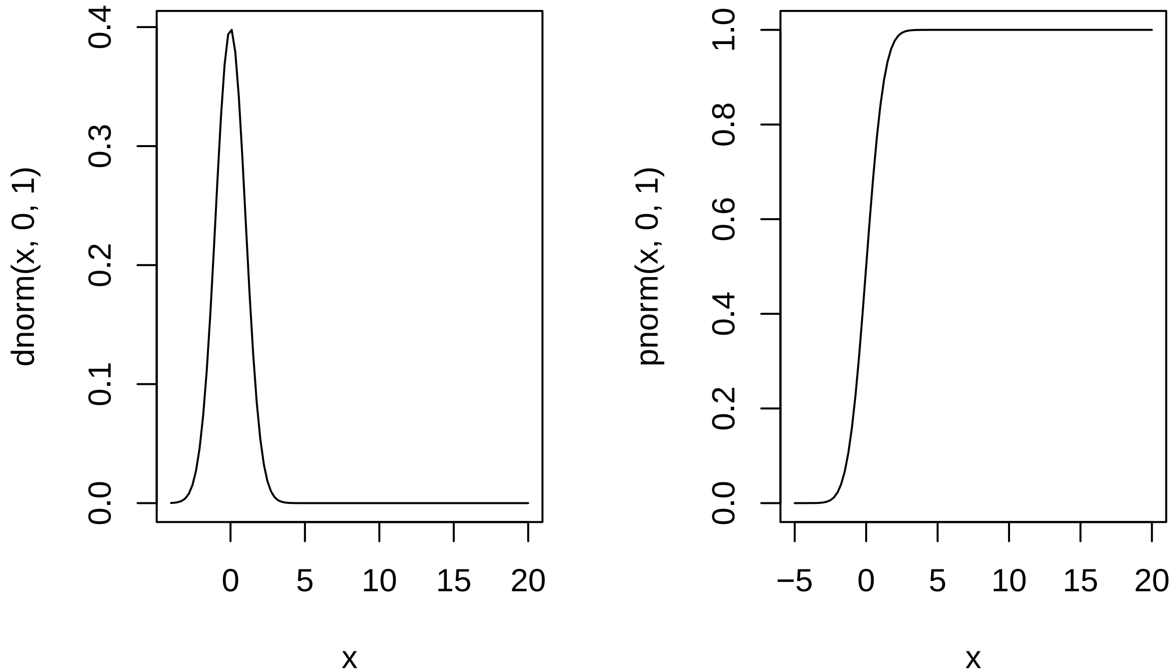

```r
curve(dnorm(x,0,1))
```



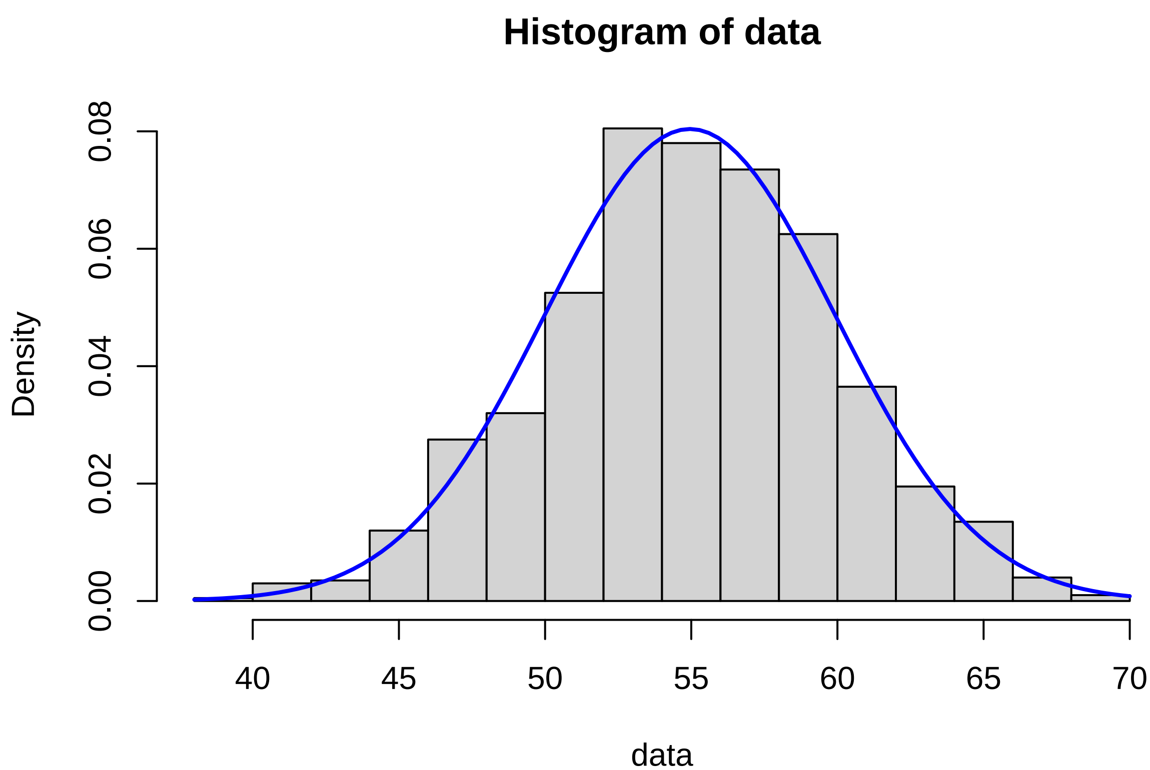

```
curve(dnorm(x,0,1),-3,3)
```




```
par(mfrow=c(1,2))
curve(dnorm(x,0,1),-4,20)
curve(pnorm(x,0,1),-5,20)
```

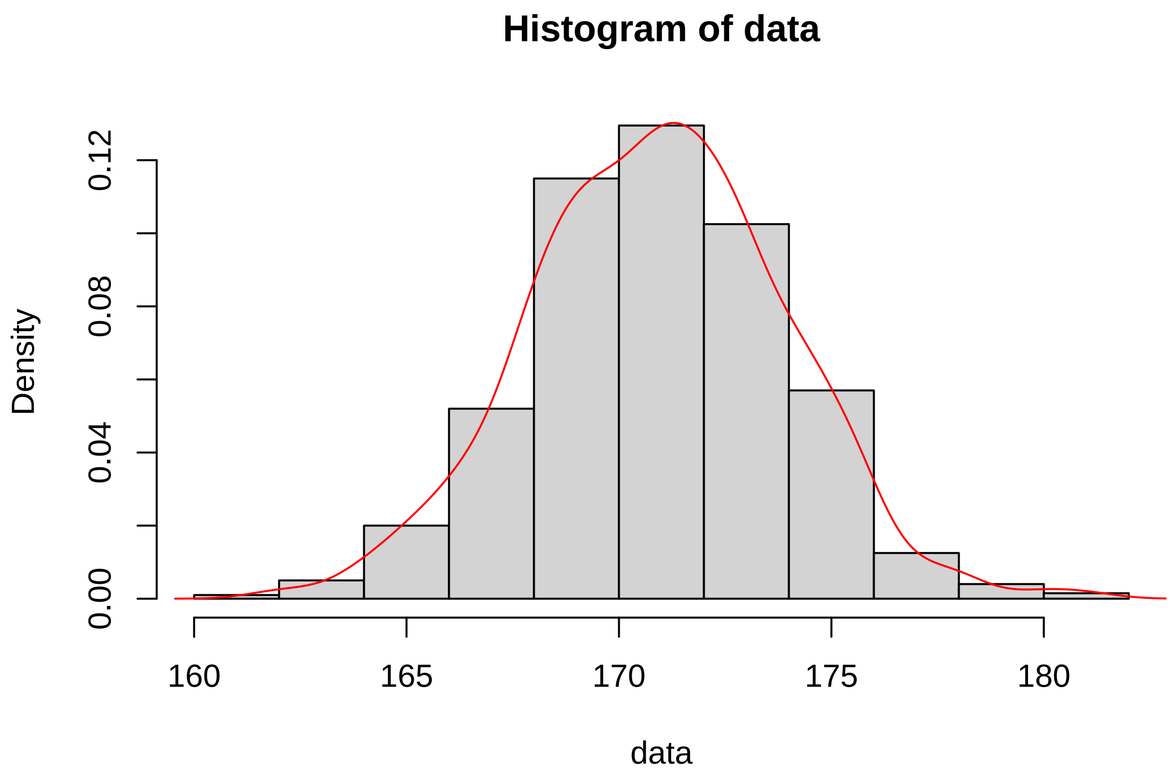



```
par(mfrow=c(1,1))
data=rnorm(1000, 55, 5)
hist(data, probability = T)
curve(dnorm(x, mean = mean(data), sd = sd(data)),
      col = "blue", lwd = 2, add = TRUE)
```

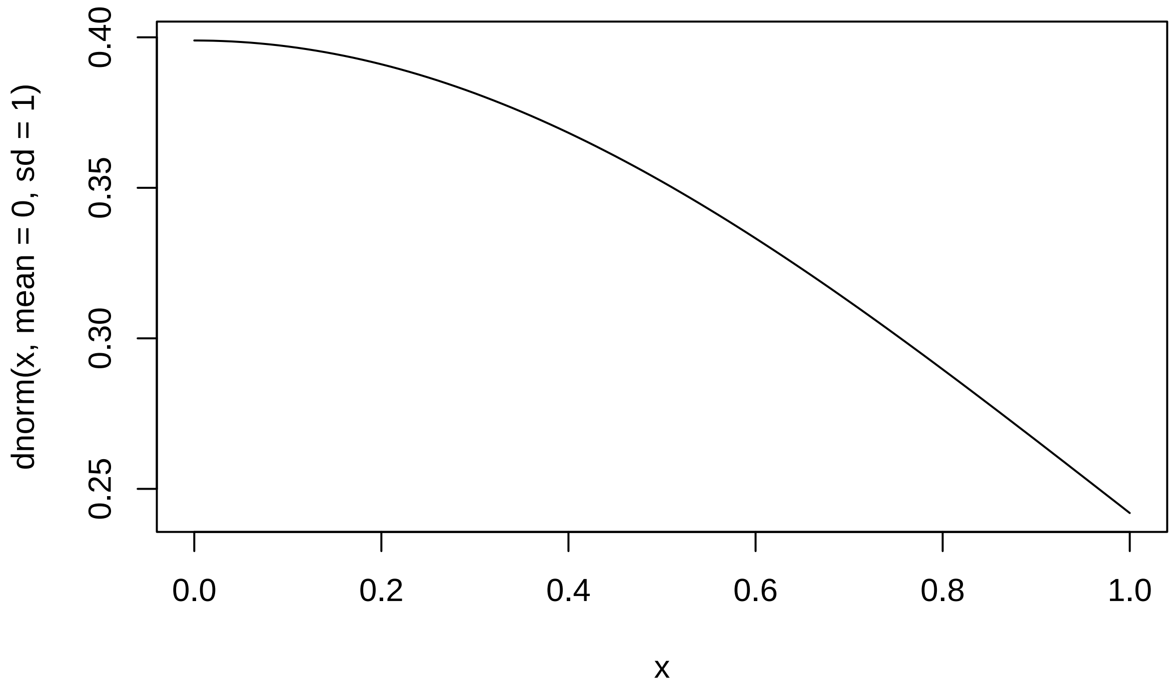



```
#####
```

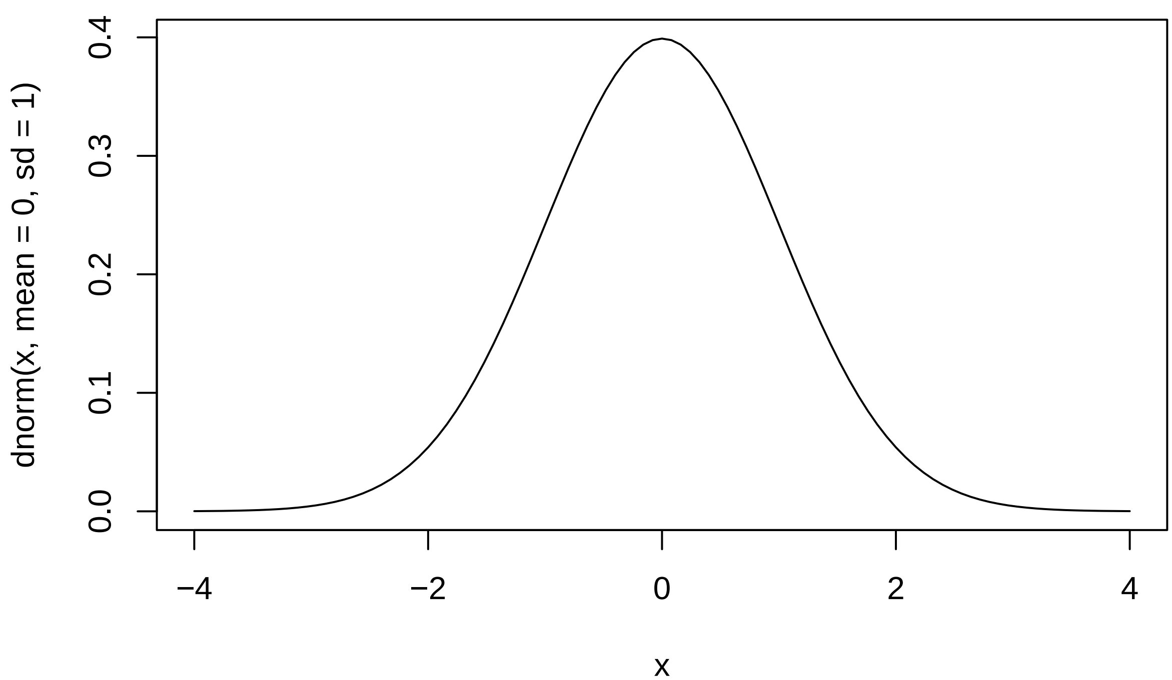
```r
data <- rnorm(1000, mean = 171, sd = 3)
hist(data, probability = T)
lines(density(data), col='red')
```



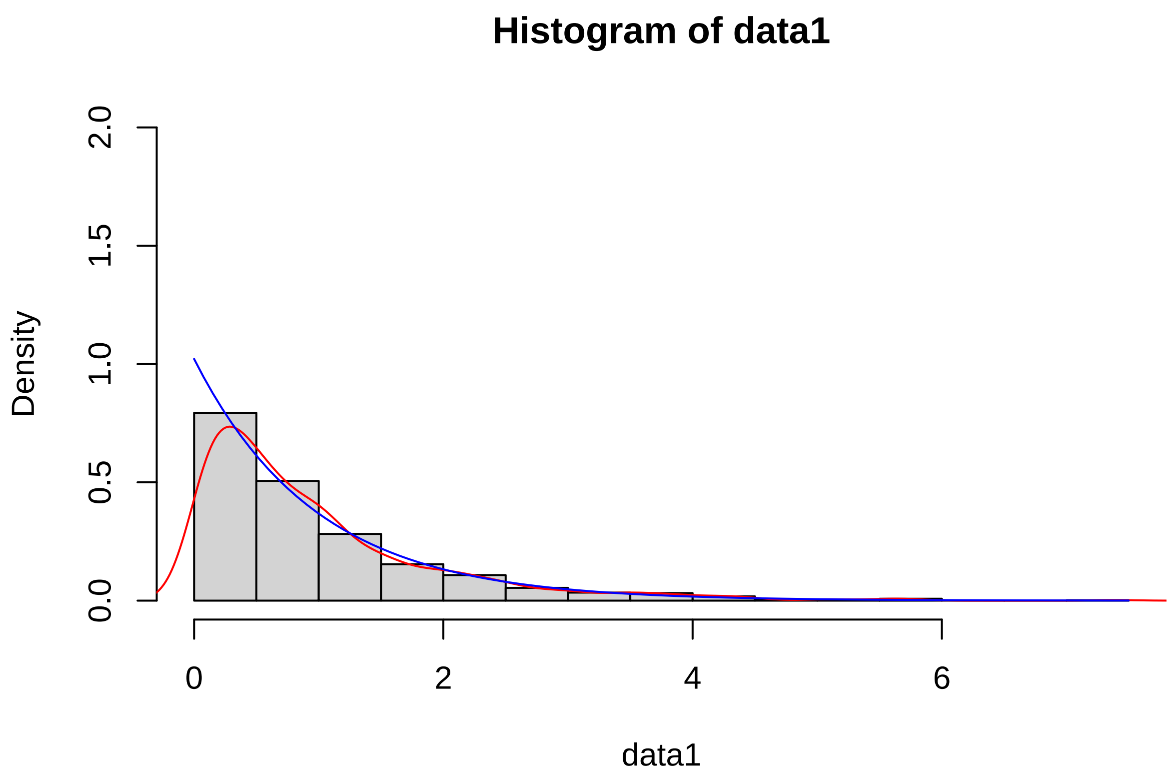

```r
par(mfrow=c(1,1))


curve(dnorm(x, mean = 0, sd = 1))
```




```r
curve(dnorm(x, mean = 0, sd = 1), from = -4, to = 4)
```




```r
data1<- rexp(1000, 1)
hist(data1, probability = T, ylim=c(0,2))
lines(density(data1), col='red')
curve(dexp(x, rate = 1/mean(data1)), add = TRUE, col='blue')
```




```r
data2<- rpois(100, 3)
hist(data2, probability = T)
lines(density(data2), col='red')
curve(dpois(x, lambda = mean(data2)),
      add = TRUE, col='blue')
```

```
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.100000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.200000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.300000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.400000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.500000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.600000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.700000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.800000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 0.900000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 1.100000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 1.200000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 1.300000

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## Warning in dpois(x, lambda = mean(data2)): non-integer x = 1.700000
```

```
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 1.800000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 1.900000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 2.100000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 2.200000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 2.300000
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## Warning in dpois(x, lambda = mean(data2)): non-integer x = 2.700000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 2.800000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 2.900000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.100000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.200000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.300000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.400000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.500000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.600000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.700000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.800000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 3.900000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.100000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.200000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.300000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.400000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.500000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.600000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.700000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.800000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 4.900000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.100000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.200000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.300000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.400000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.500000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.600000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.700000
```

```
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.800000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 5.900000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 6.100000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 6.200000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 6.300000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 6.400000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 6.500000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 6.600000
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## Warning in dpois(x, lambda = mean(data2)): non-integer x = 6.800000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 6.900000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.100000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.200000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.300000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.400000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.500000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.600000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.700000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.800000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 7.900000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.100000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.200000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.300000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.400000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.500000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.600000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.700000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.800000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 8.900000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.100000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.200000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.300000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.400000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.500000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.600000
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.700000
```
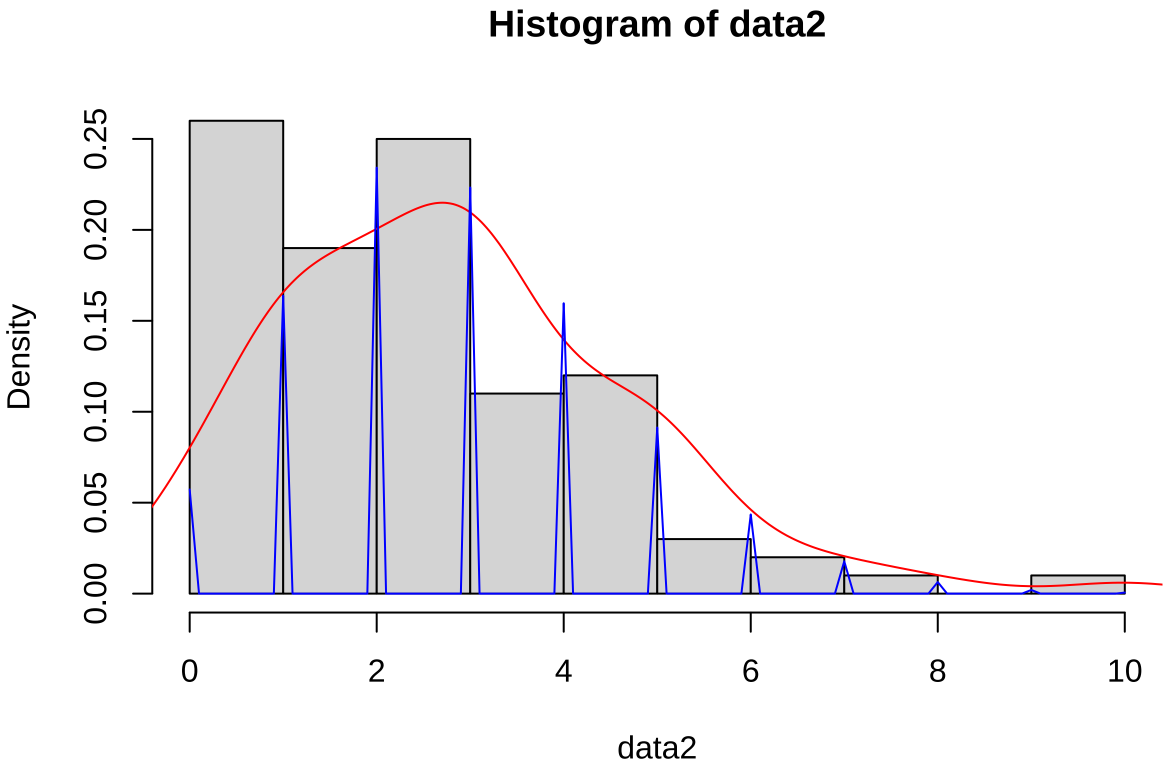
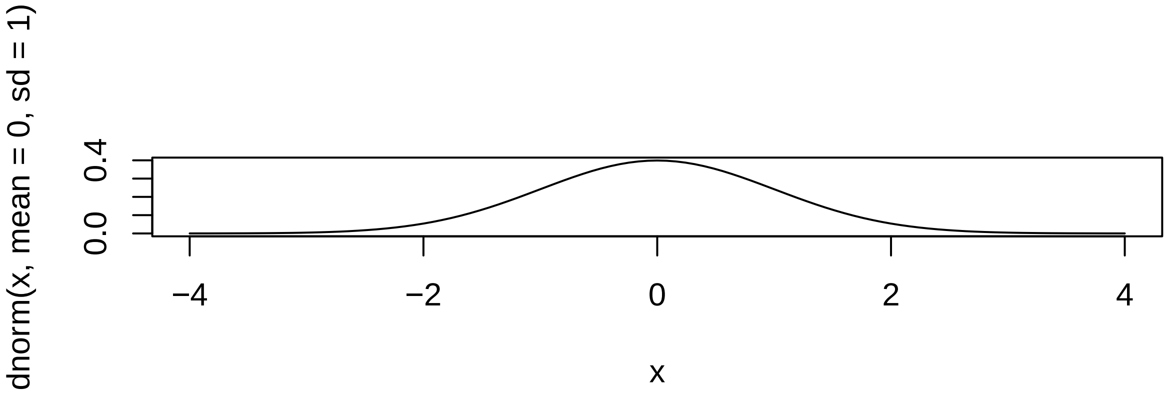
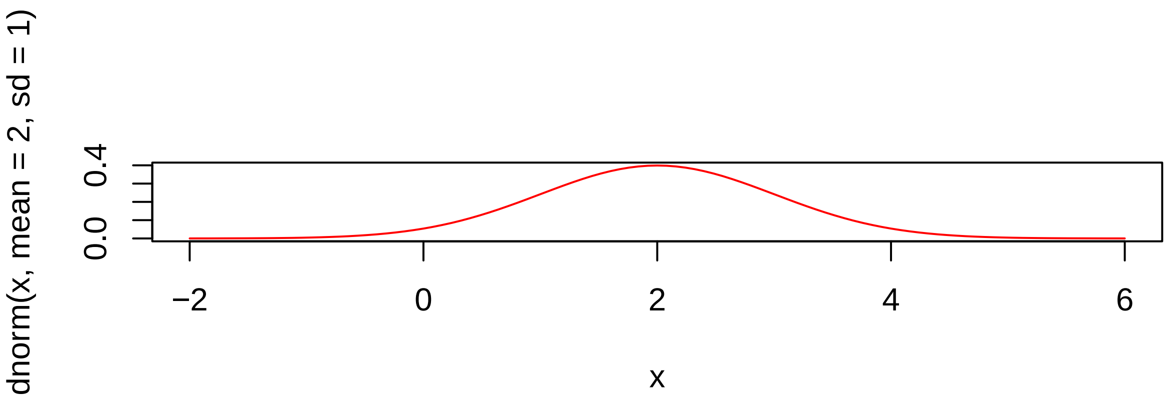

```
## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.800000

## Warning in dpois(x, lambda = mean(data2)): non-integer x = 9.900000
```

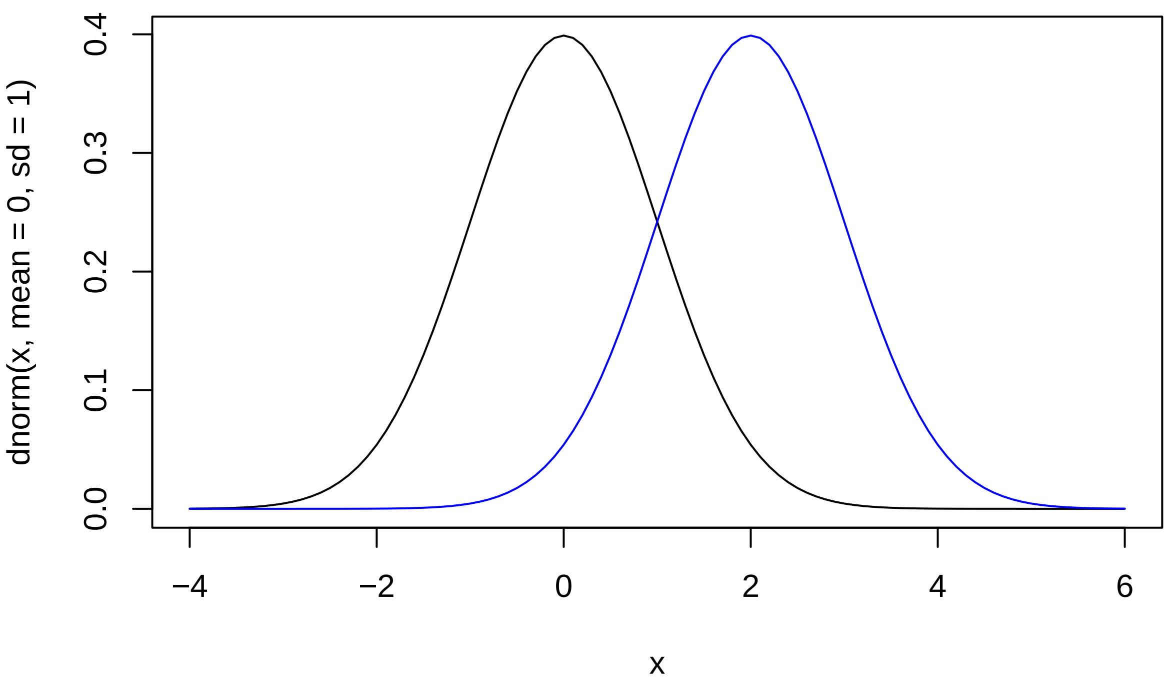



```r
par(mfrow=c(2,1))
curve(dnorm(x, mean = 0, sd = 1),  -4,  4)
curve(dnorm(x, mean = 2, sd = 1),  -2, 6, col='red')
```

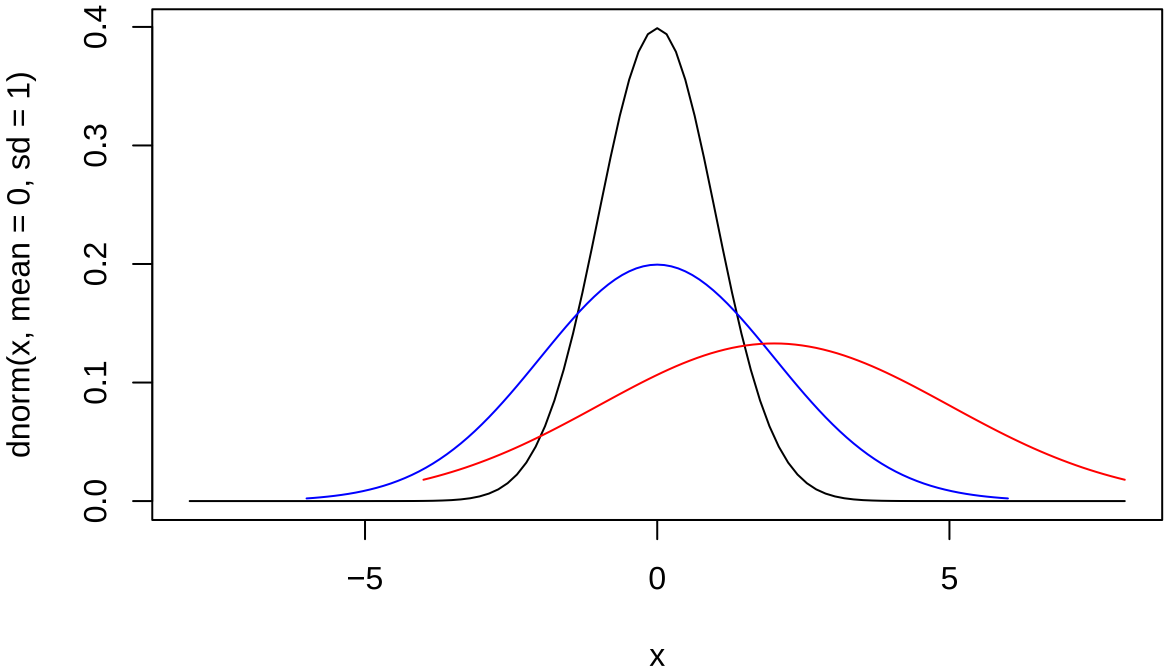




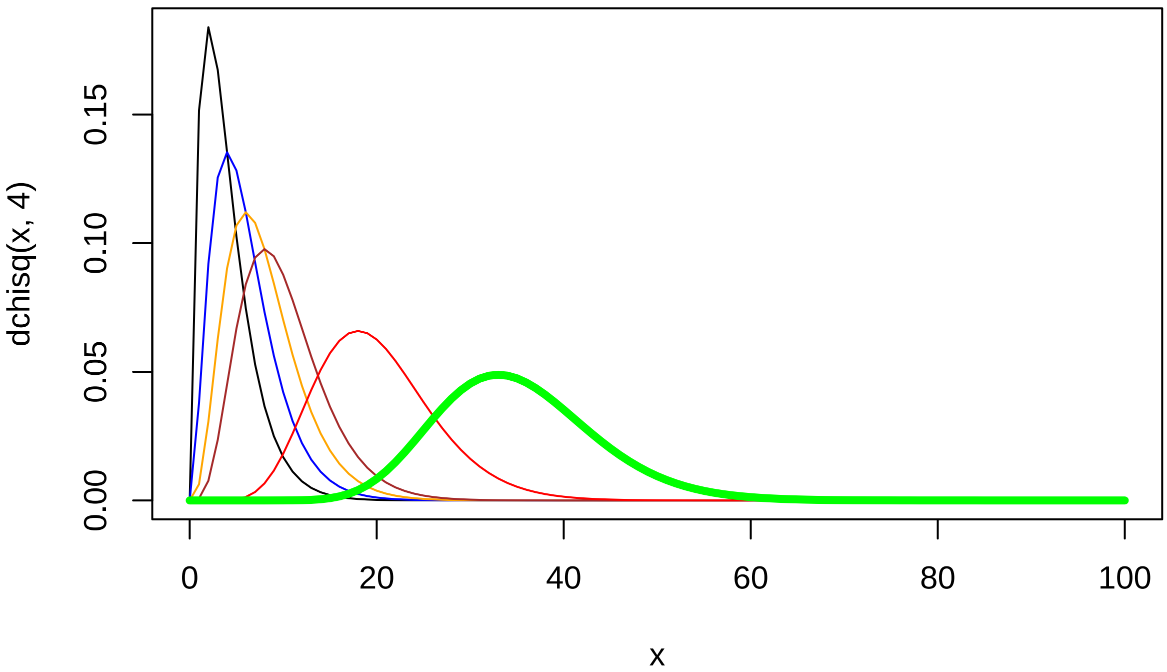

```r
#par(mar=c(1,1,1,1))

par(mfrow=c(1,1))
curve(dnorm(x, mean = 0, sd = 1),  -4,  6)
curve(dnorm(x, mean = 2, sd = 1),  -4,  6,
      col='blue', add=T)
```



```r
curve(dnorm(x, mean = 0, sd = 1),  -8,  8)
curve(dnorm(x, mean = 0, sd = 2),  -6,  6,
      col='blue', add=T)
curve(dnorm(x, mean = 2, sd = 3),  -4,  8,
      col='red', add=T)
```




```r
curve(dchisq(x, 4),  0,  100)
curve(dchisq(x, 6),  0,  100,
      add = T, col='blue')
curve(dchisq(x, 8),  0,  100,
      add = T, col='orange')
curve(dchisq(x, 10),  0,  100,
      add = T, col='brown')
curve(dchisq(x, 20),  0,  100,
      add = T, col='red')
curve(dchisq(x, 35),  0,  100,
      add = T, col='green',
      lwd=4)
```


dchisq(x, 4)
0.00
0.05
0.10
0.15
0
20
40
60
80
100
x